



Infrastructure Finance

Volatility Spillovers, Factor Structure, and Portfolio Optimization of Indian Infrastructure Assets

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Master's Thesis Project II

INTRODUCTION

Motivation & Context

Infrastructure as an Emerging Asset Class

- Offers stable cash flows, long-duration contracts, diversification potential
- In developed markets: evolved into equity-like tradable assets (REITs)
- Transition: segmented to integrated markets

Core Problem

- Behaviour of Indian infrastructure assets is not well understood
- Developed market evidence may not apply to India

Indian Context: Early-Stage Market

- Recent introduction of:
 - InvITs (infrastructure assets)
 - REITs (real estate assets)
- Structural differences:
 - InvITs : regulated, cash-flow driven
 - REITs: property market exposure
 - Infra equities: macro-driven stocks

Key Uncertainty

- Are these assets:
 - Globally integrated like developed markets?
 - Or segmented, illiquid, and idiosyncratic?

INTRODUCTION

Existing Gap, Approach & Contributions

Why Existing Models May Fail

- Standard asset pricing assumes:
 - Returns driven by systematic factors
- But in emerging markets:
 - Illiquidity, delayed price discovery, frictions dominate
- Traditional models may have low explanatory power

Methodological Approach (4 Phases)

- Phase 1: Market integration (spillovers)
- Phase 2: Factor drivers (macro + liquidity)
- Phase 3: ML prediction (Bayesian + priors)
- Phase 4: Portfolio performance

Key Questions

1. Are Indian infrastructure assets globally integrated or segmented?
2. What drives their returns: macro factors or idiosyncratic forces?
3. Can returns be predicted using ML + economic structure?
4. Do predictions create real portfolio value?

Main Contributions

- Evidence of low global integration (segmentation)
- Weak factor explanation for InvITs (liquidity-driven)
- Improved prediction using hierarchical ML
- Portfolio gains depend on signal processing

RELATED RESEARCH

1. Behaviour of Securitized Real Assets

- REITs have evolved from bond-like to equity-like instruments, indicating increasing market integration (Glascock et al., 2000)
- Their returns reflect a combination of equity, real estate, and idiosyncratic factors, rather than a single asset class (Clayton & MacKinnon, 2003)
- The strength of these linkages varies with market conditions and data frequency, highlighting partial integration (Cotter & Stevenson, 2006)

2. Market Integration & Spillovers

- Financial integration is time-varying and intensifies during crises, rather than being constant (Diebold & Yilmaz, 2012)
- Developed REIT markets tend to act as net transmitters of shocks to other markets (Antonakakis et al., 2021)
- Emerging markets exhibit lower and less stable integration, especially in early stages (Pham, 2012)

3. Modelling, Prediction & Portfolio Approaches

- Volatility and correlations are dynamic and asymmetric, requiring advanced models such as DCC-GARCH (Engle, 2002; Cappiello et al., 2006)
- Machine learning models can capture complex, nonlinear relationships and improve return prediction (Gu et al., 2020)
- Portfolio optimisation methods like HRP provide more stable allocations in small-sample settings (López de Prado, 2016)

4. Research Gap

- Existing literature is largely focused on developed REIT markets, with limited evidence for India
- The behaviour of InvITs and infrastructure assets remains underexplored
- No study jointly examines integration, return drivers, predictability, and portfolio performance

RESEARCH OBJECTIVES

Motivation

- Indian InvITs and REITs are rapidly growing, but their behaviour is not well understood
- Existing evidence focuses on developed markets, with limited insight for India

Central Research Question

Are Indian InvIT and REIT markets segmented from global markets, and if so:

- What drives their returns?
- Can these returns be predicted?
- Do predictions create real portfolio value?

Four-Phase Research Framework

Phase 1: Market Integration (Analyze volatility, correlations, and spillovers to test segmentation)

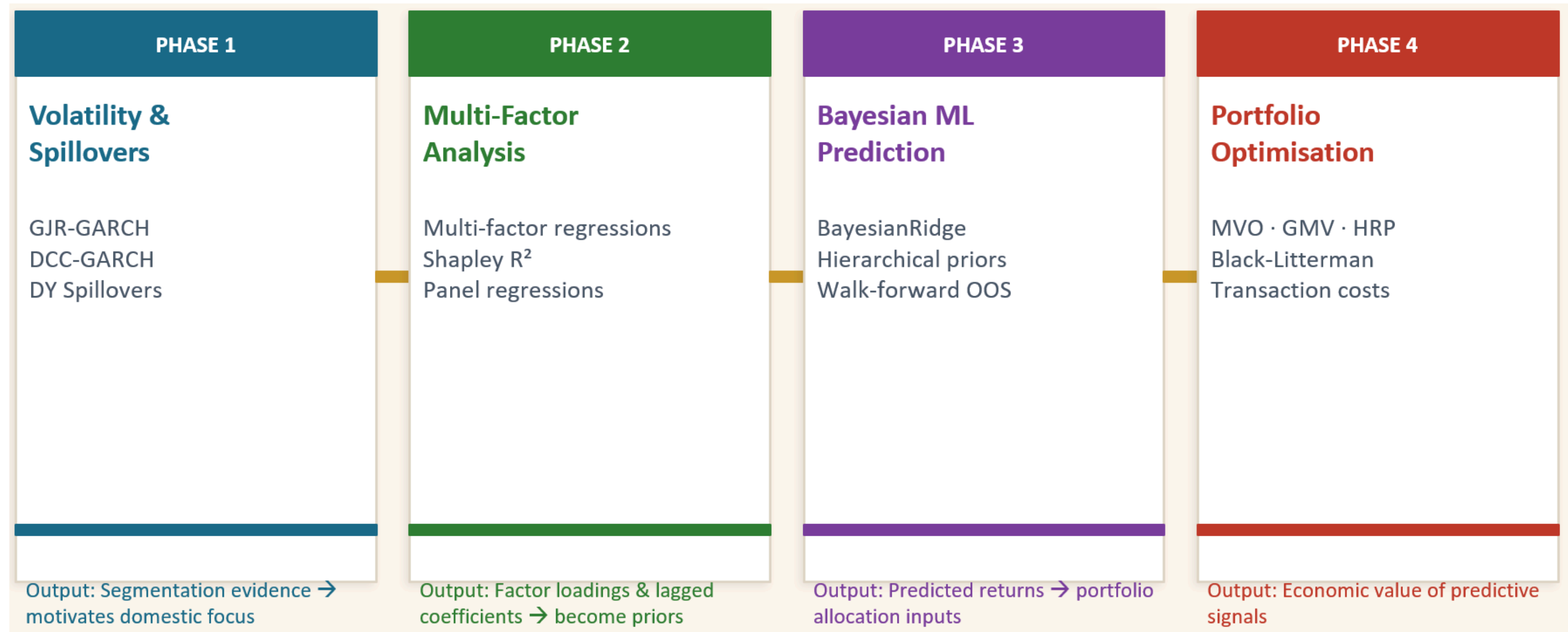
Phase 2: Return Drivers (Identify macroeconomic and liquidity factors affecting returns)

Phase 3: Predictability (Test out-of-sample return prediction using ML with economic priors)

Phase 4: Portfolio Value (Evaluate whether predictions improve investment performance)

RESEARCH OBJECTIVES

4 Phase analysis



Note: GJR-GARCH: Glosten-Jagannathan-Runkle Generalized Autoregressive Conditional Heteroskedasticity; DCC-GARCH: Dynamic Conditional Correlation GARCH; FEVD: Forecast Error Variance Decomposition; MVO: Mean-Variance Optimization; GMV: Global Minimum Variance; HRP: Hierarchical Risk Parity.

RESEARCH OBJECTIVES

Tickers used

<i>Category</i>	<i>Ticker</i>	<i>Name</i>	<i>Category</i>	<i>Ticker</i>	<i>Name</i>	<i>Category</i>	<i>Ticker</i>	<i>Name</i>
India InvIT	IRBINVIT.NS	IRB InvIT Fund	US REIT	AMT	American Tower Corp	Europe Stock	FER.MC	Ferrovial (Madrid)
India InvIT	INDIGRID.NS	IndiGrid InvIT		CCI	Crown Castle Inc	Europe Stock	NG.L	National Grid (London)
India InvIT	PGINVIT.NS	PowerGrid InvIT		EQIX	Equinix	Europe Stock	ENGI.PA	Engie (Paris)
India REIT	EMBASSY.N	Embassy Office Parks REIT		DLR	Digital Realty Trust	Europe Stock	RWE.DE	RWE AG (Frankfurt)
India REIT	MINDSPACE.	Mindspace Business Parks	US MLP	EPD	Enterprise Products Partners	Europe REIT	CLNX.MC	Cellnex Telecom (Madrid)
India REIT	BIRET.NS	Brookfield India REIT				Europe Market	VGK	Vanguard FTSE Europe ETF
India Stock	IRB.NS	IRB Infrastructure Developers	US MLP	ET	Energy Transfer LP			
India Stock	LT.NS	Larsen & Toubro	US MLP	MPLX	MPLX LP	Table 3		
India Stock	ADANIPORT	Adani Ports & SEZ	US Stock	BIP	Brookfield Infrastructure Partners			
India Stock	NTPC.NS	NTPC Limited						
India Stock	POWERGRID	Power Grid Corporation of	US Stock	NEE	NextEra Energy			
India Market	^NSEI	Nifty 50	US Stock	DUK	Duke Energy			
India Market	^CNXINFRA	Nifty Infrastructure Index	US Market	IFRA	iShares U.S. Infrastructure ETF			
India Market	INR=X	USD/INR Exchange Rate						
India Market	(manual	India 10-Year Government						

Table 1

Table 2

PHASE 1: VOLATILITY DYNAMICS & SPILLOVER ANALYSIS

METHODOLOGY

Objective: Measure the degree of integration between Indian infrastructure assets and global markets.

Step 1: Volatility Modelling

- GJR-GARCH(1,1) used to model conditional volatility
- Captures: Volatility clustering and Asymmetric response to negative shocks

Step 2: Dynamic Correlations

- DCC-GARCH (Engle, 2002) used to estimate time-varying correlations
- Measures how co-movement between markets evolves over time

GARCH for Individual volatility
DCC for Market co-movement
FEVD for Shock propagation

Step 3 Spillover Analysis

- VAR + Generalized FEVD (Diebold & Yilmaz, 2012)
- Quantifies: Cross-market shock transmission, Net transmitters vs receivers, and Total system connectedness

PHASE 1: VOLATILITY DYNAMICS & SPILLOVER ANALYSIS

Results

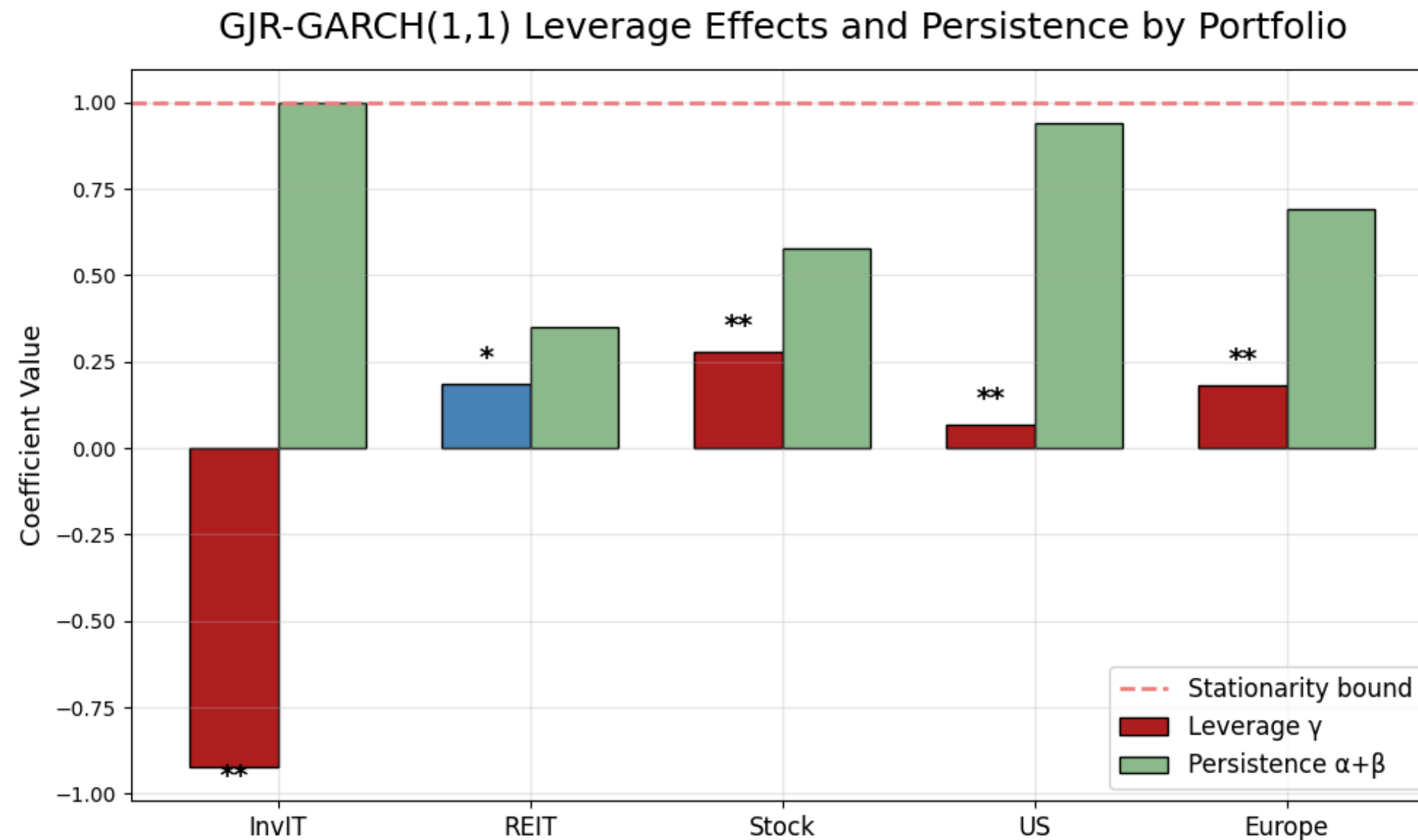


Fig.1

- Significant leverage effects observed for:
 - India_Stock
 - US_Infra
 - Europe_Infra
- India_REIT shows weak asymmetry; InvIT leverage estimate unstable due to non-stationarity
- US_Infra exhibits highest volatility persistence ($\alpha + \beta = 0.94$; half-life ≈ 11 days)
- Indian REITs display rapid mean reversion (half-life < 1 day)

Indian InvITs and REITs exhibit volatility behaviour different from mature global infrastructure markets

PHASE 1: VOLATILITY DYNAMICS & SPILLOVER ANALYSIS

Results

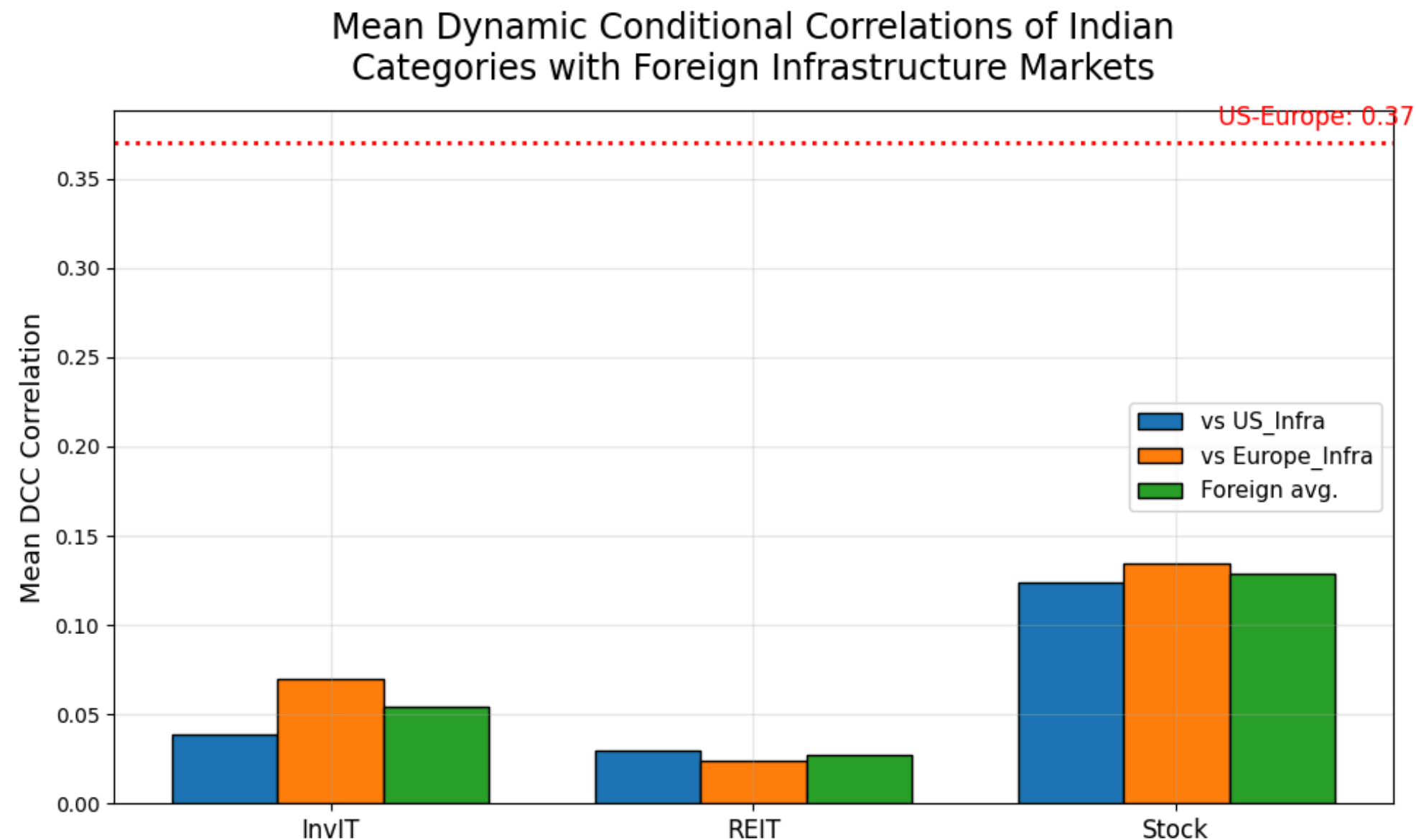


Fig.2

- Indian infrastructure assets show very low correlation with global infrastructure markets
- Mean correlations:
 - InvIT-US: 0.04
 - REIT-US: 0.03
 - Stock-US: 0.12
- These are far below:
 - US-Europe correlation: 0.37
- Domestic Indian correlations are stronger than international linkages

Indian infrastructure markets remain weakly integrated with global infrastructure systems

PHASE 1: VOLATILITY DYNAMICS & SPILLOVER ANALYSIS

Results

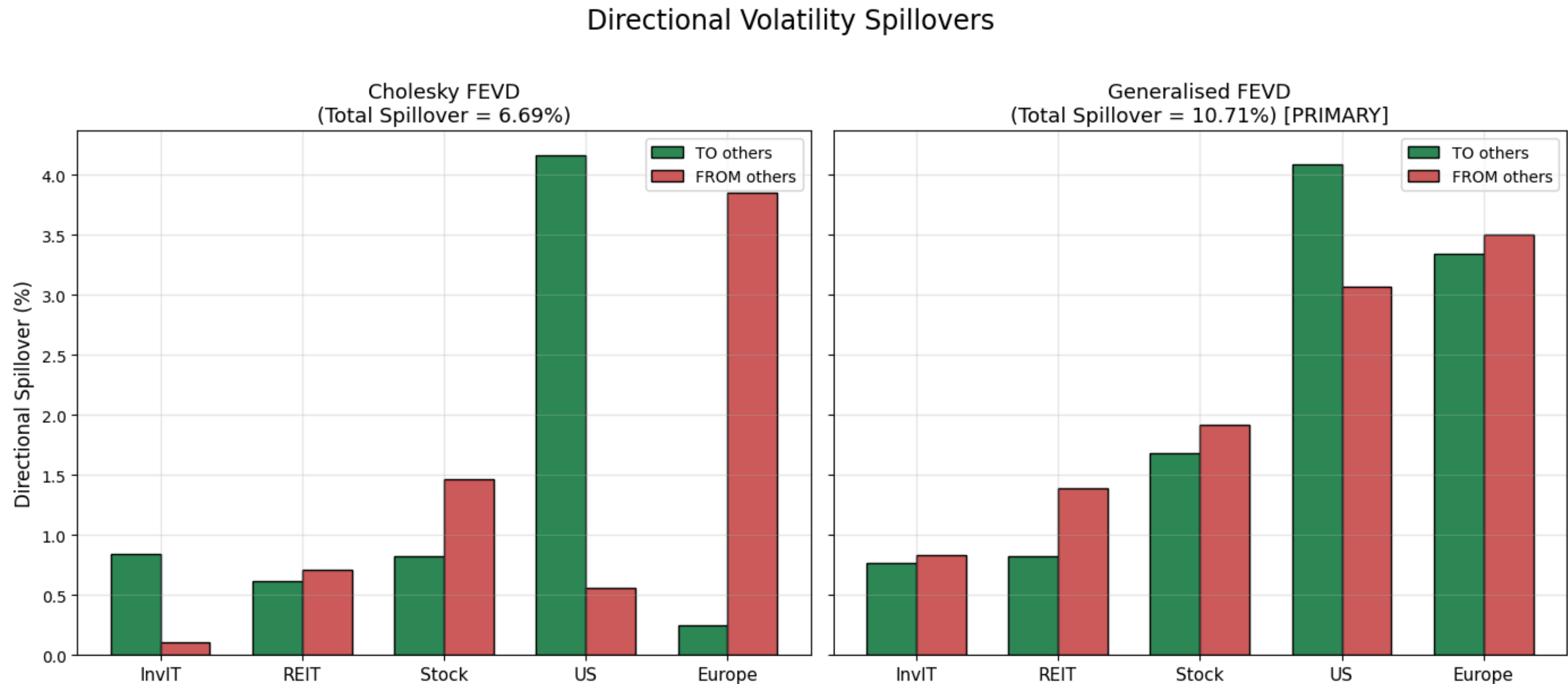


Fig.3

Indian infrastructure assets are strongly segmented from global infrastructure markets

PHASE 2: MULTI-FACTOR ANALYSIS OF DOMESTIC RETURN DRIVERS

Methodology

Objective: To examine whether the weakly integrated behaviour identified in Phase 1 can be explained by domestic macro-financial and liquidity factors

- Step 1 [**Factor Construction**]: Market, macroeconomic, sectoral, volatility, and liquidity factors are constructed using weekly log returns and changes in economic variables.
- Step 2 [**Multi-Factor Regression**]: OLS regressions with Newey-West HAC standard errors are used to estimate factor sensitivities and ensure robust statistical inference.
- Step 3 [**Variance Attribution**]: Shapley decomposition is applied to measure the relative contribution of each factor to total explanatory power (R^2).
- Step 4 [**Cross-Category Testing**]: Panel regressions with interaction terms test whether InvITs, REITs, and infrastructure stocks respond differently to the same factors.
- Step 5 [**Predictive Extension**]: Rolling and lagged regressions evaluate time-varying exposures and delayed price discovery effects.

Identify domestic drivers of returns & determine whether economically meaningful predictive structure exists

PHASE 2: MULTI-FACTOR ANALYSIS OF DOMESTIC RETURN DRIVERS

Results: Explanatory Power Across Models

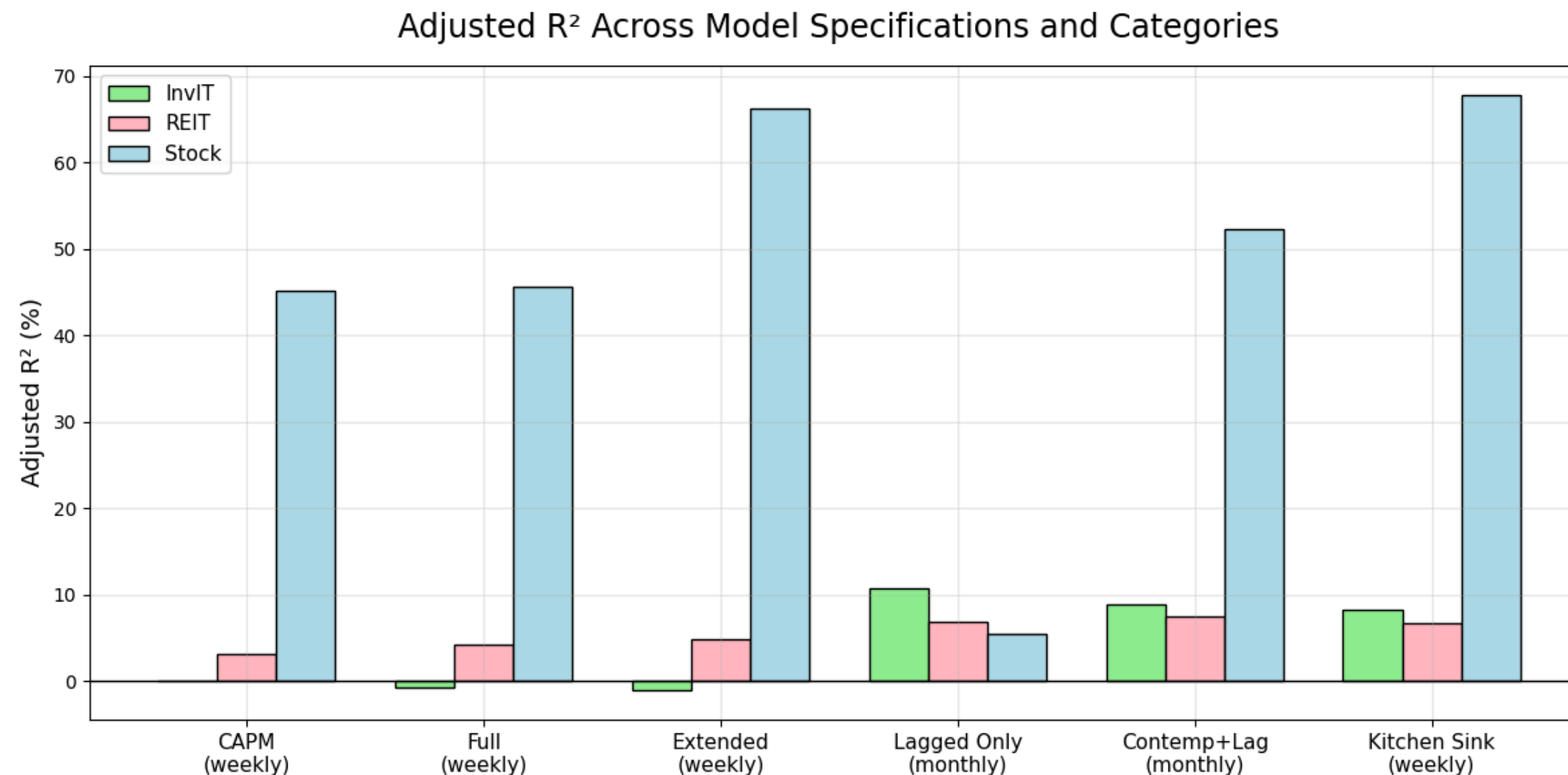


Fig.4

- Infrastructure stocks are strongly explained by macro-financial factors (*Adjusted R² up to 67.8%*)
- REITs exhibit partial explainability (*maximum Adjusted R² around 7.5%*)
- InvITs remain largely idiosyncratic under weekly models
- Monthly lagged models substantially improve InvIT explainability (*Adjusted R² rises to 10.8%*)

Delayed adjustment effects are strongest in InvITs, consistent with lower liquidity and slower price discovery

PHASE 2: MULTI-FACTOR ANALYSIS OF DOMESTIC RETURN DRIVERS

Results: Factor Loadings & Economic Interpretation

<i>Factor</i>	<i>InvIT</i>	<i>REIT</i>	<i>Stock</i>
Nifty50	Weak	Moderate	Strong
Yield10Y	Weak	Weak	Positive
INR/USD	Weak	Negative	Weak
Oil	Insignificant	Insignificant	Insignificant

Table 4

- Stock returns are highly sensitive to equity markets ($\beta = 1.228$ for Nifty50)
- REITs show moderate market exposure and currency sensitivity
- InvITs exhibit near-zero market beta and weak macro linkage
- Oil prices do not significantly explain returns

InvITs behave differently from traditional equity-linked infrastructure assets

PHASE 2: MULTI-FACTOR ANALYSIS OF DOMESTIC RETURN DRIVERS

Results: Shapley Variance Decomposition

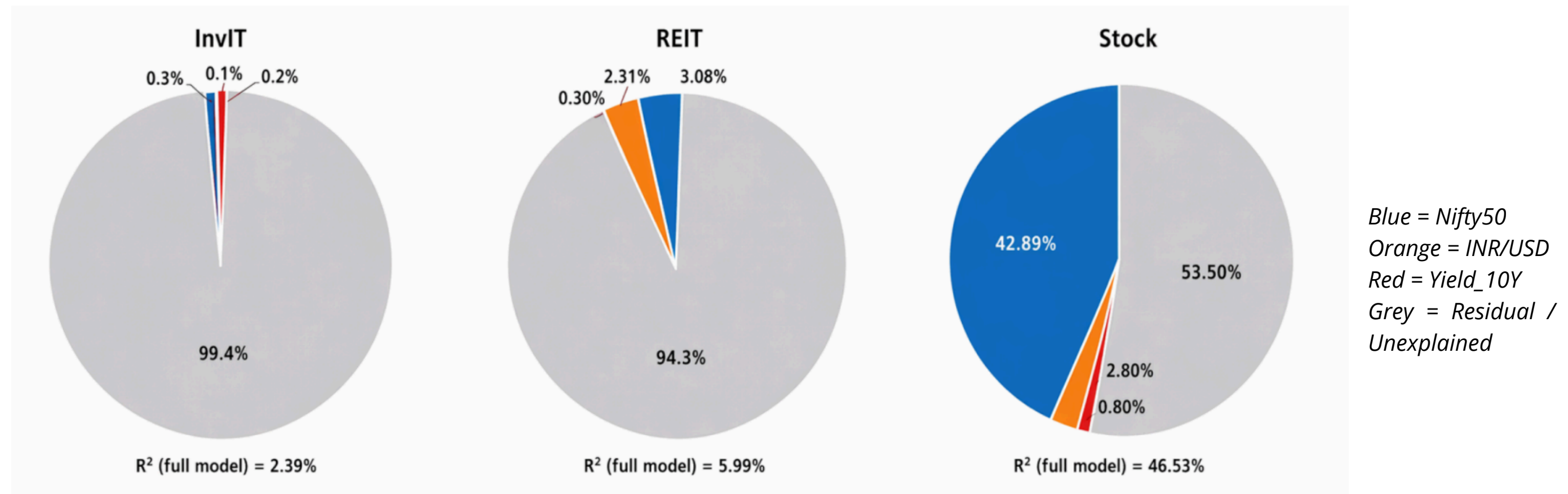


Fig.5

- Nifty50 explains most of Stock return variation (*around 43% of R^2*)
- REIT explanatory power is split across: market, currency & idiosyncratic effects
- InvIT variance remains overwhelmingly unexplained (*around 99% idiosyncratic*)

Traditional macro factors have very limited ability to explain InvIT returns

PHASE 2: MULTI-FACTOR ANALYSIS OF DOMESTIC RETURN DRIVERS

Results: Cross-Category Interaction Tests

<i>Interaction</i>	<i>Coefficient</i>	<i>Interpretation</i>
Nifty50 × InvIT	-1.198	Much lower market beta
Nifty50 × REIT	-1.082	Lower market exposure
Yield × InvIT	-0.075	Higher rate sensitivity

Table 5

- InvIT and REIT sensitivities differ significantly from Stocks
- InvITs exhibit:
 - very low equity-market exposure
 - stronger interest-rate sensitivity
- Results remain robust under clustered and HAC errors

Infrastructure asset categories cannot be treated as homogeneous equity instruments

PHASE 2: MULTI-FACTOR ANALYSIS OF DOMESTIC RETURN DRIVERS

Results: Lagged Models

<i>Predictor</i>	<i>InvIT</i>	<i>REIT</i>	<i>Stock</i>	<i>Main Interpretation</i>
Lagged Market Return (Nifty50 _{t-1})	0.23	0.26	0.44	Common momentum effect across all categories
Lagged INR/USD	0.56	-0.22	-0.6	InvITs respond differently to currency movements
Lagged 10Y Yield	-0.11	0.09	0.1	REITs show sensitivity to interest-rate changes
Illiquidity (Amihud)	-0.48	0	0	Liquidity effects are important only for InvITs
Model Adjusted R ²	10.80%	6.80%	5.50%	Monthly lagged factors improve predictability

Table 6

- Lagged Nifty50 is significant across all categories
- Predictive effects emerge more strongly at monthly frequency
- InvITs show sensitivity to:
 - lagged currency movements
 - illiquidity (Amihud)
- These coefficients are used as Bayesian priors in Phase 3

Phase 2 identifies economically meaningful predictive structure for machine learning models

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Methodology: Bayesian Predictive Framework

Objective:

- Test whether weak but structured signals from Phase 2 can generate out-of-sample predictability
- Move from explanatory modelling to predictive machine learning

Dataset Construction:

- Monthly pooled panel across: (i) InvIT, (ii) REIT & (iii) Infrastructure Stocks
- around 170 pooled observations to improve statistical power
- Features include lagged macro variables, yield, autoregressive terms, and illiquidity measures

Predictive Design:

- Expanding-window walk-forward framework
- Model retrained sequentially using only past information
- Prevents lookahead bias and mimics real forecasting conditions

Core Model:

- Bayesian Ridge Regression with coefficient shrinkage
- Designed for low signal-to-noise financial data
- Combines regularisation with probabilistic inference

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Methodology: Hierarchical Priors & Evaluation

Hierarchical Bayesian Structure:

- Phase 2 factor coefficients used as economically informed priors
- Priors blended with data-driven estimates
- Prior weights selected dynamically using LOOCV (Leave-One-Out Cross-Validation)

Key Advantage:

- Combines:
 - cross-sectional learning from pooled data
 - category-specific economic structure
- Prevents overfitting while preserving interpretability

Evaluation Metrics

- Out-of-sample R^2
- Information Coefficient (IC)
- Directional Accuracy
- Diebold–Mariano predictive comparison test

Out-of-sample predictions generated in Phase 3 become the portfolio signals used in Phase 4 optimisation & backtesting

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Results: Out-of-Sample Predictive Performance

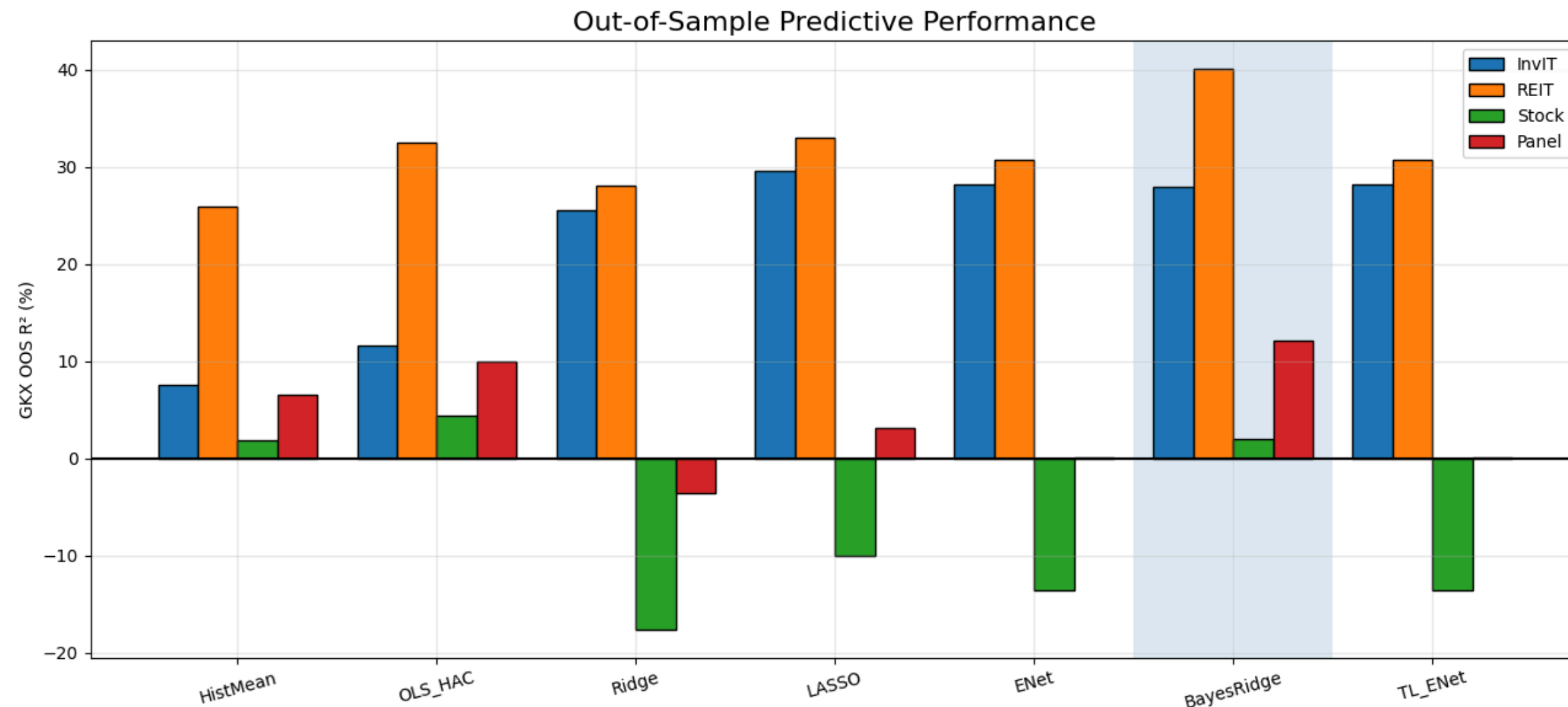


Fig.6

- BayesRidge is the best overall model across categories
- Only model with: positive GKX R2 for InvIT, REIT, Stock, and Panel; positive Campbell-Thompson panel R2
- REIT prediction performance is exceptionally strong: GKX R2=40.05%
- LASSO and ENet perform well for InvIT/REIT but fail badly for Stocks
- Informative priors from Phase 2 improve stability and prediction quality

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Results: Why BayesRidge Wins

<i>Model</i>	<i>Panel GKX R2</i>	<i>Panel CT R2</i>	<i>Panel IC</i>
HistMean	6.52%	-5.37%	-0.026
OLS_HAC	9.98%	-1.47%	-0.202
LASSO	3.07%	-9.26%	0.052
ENet	0.02%	-12.70%	0.011
BayesRidge	12.09%	0.91%	0.254

Table 7

- Hierarchical priors stabilize estimation
- Small sample size favors shrinkage models
- LOOCV-selected blend weight avoids overfitting
- Smooth coefficient shrinkage works better than aggressive sparsity
- REIT trend persistence partially captured

The model benefits more from stable coefficient estimation than from aggressive feature selection

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Results: Directional Accuracy & True Predictive Skill

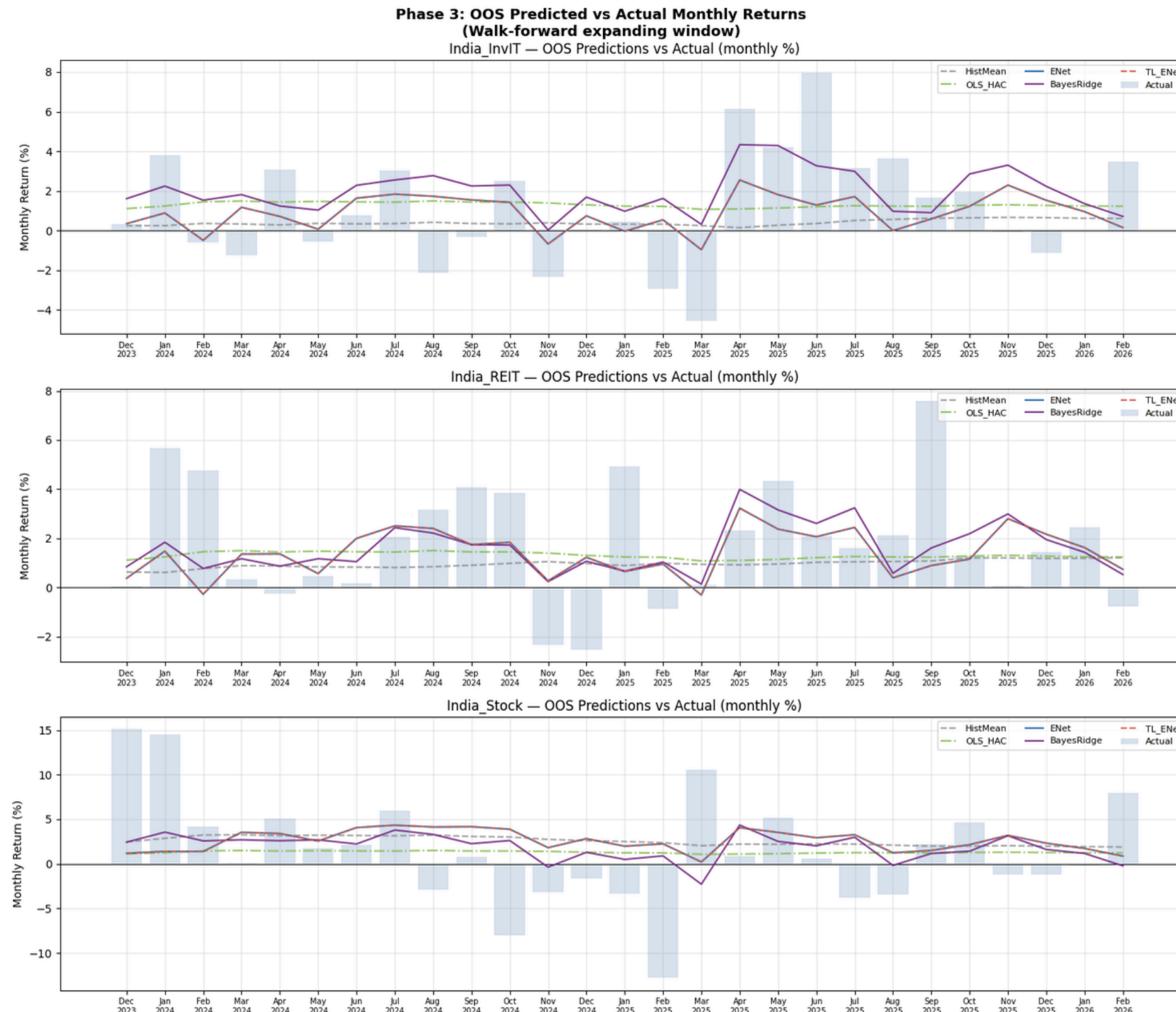


Fig.7

- REIT sample is highly imbalanced:
 - 22 positive months
 - only 5 negative months
- Constant “predict-up” strategy already achieves high accuracy
- BayesRidge fails to predict downturns
- Therefore:
 - Directional Accuracy (DA) is not reliable here
 - Information Coefficient (IC) is more meaningful

Key Findings:

- BayesRidge Panel IC = 0.254
- REIT IC = 0.360
- Indicates strong ranking ability despite weak directional timing

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Results: Feature Importance via SHAP Analysis

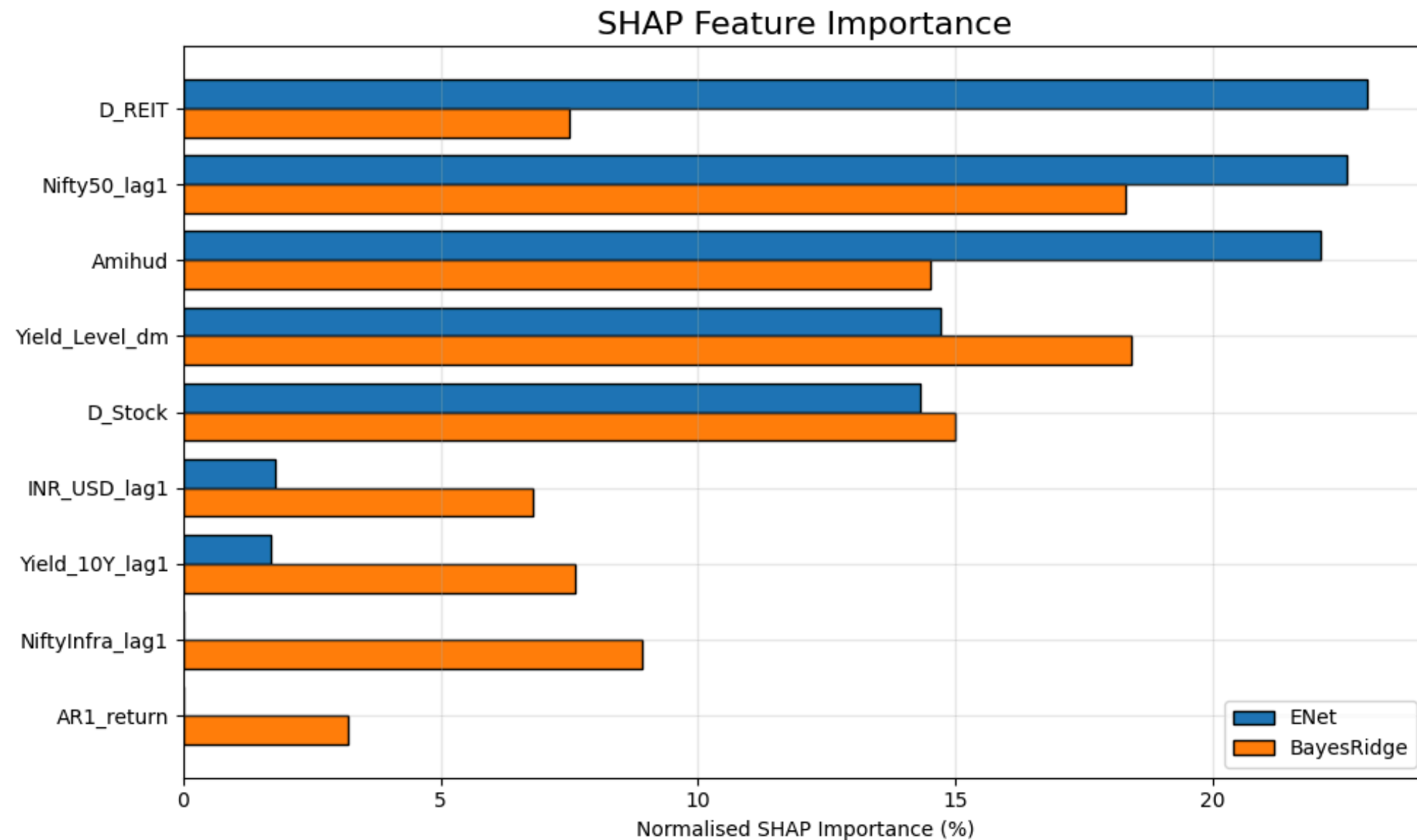


Fig.8

- Most important predictive features:
 - Nifty50_lag1
 - Amihud
 - Yield_Level_dm
- ElasticNet heavily emphasizes REIT dummy
- BayesRidge distributes importance more smoothly
- Liquidity (Amihud) remains economically important
- Market factor persistence is strong

Phase 3 ML results are broadly consistent with Phase 2 econometric findings

PHASE 3: BAYESIAN MACHINE LEARNING PREDICTION

Results: Consistency Between Econometrics and ML

<i>Feature</i>	<i>Phase 2 Significant?</i>	<i>High SHAP Importance?</i>
Nifty50_lag1	Yes	Yes
Amihud	Yes	Yes
Yield_Level_dm	Yes	Yes
NiftyInfra_lag1	No	No
AR1_return	No	No

Table 8

- Econometric significance and ML importance align closely
- Same variables emerge across:
 - HAC regressions
 - Bayesian models
 - ElasticNet
 - SHAP decomposition
- Supports robustness of identified return drivers
- Indicates economically meaningful predictive structure

Machine learning reinforces the economic structure identified in Phase 2 rather than discovering random patterns

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Methodology: Portfolio Construction Framework

Objective:

Evaluate whether Phase 3 predictive signals generate economically meaningful portfolio performance after accounting for risk and transaction costs.

Key Features:

- Uses out-of-sample predictions from Phase 3
- Ensemble weights determined dynamically using Information Coefficient (IC)
- EWMA covariance used to capture time-varying risk
- Long-only constrained optimisation framework
- Performance evaluated net of transaction costs

Phase 3 Predictions → IC-weighted Ensemble Signals → Signal Standardization → Expected Return Estimation → EWMA Risk Estimation → Portfolio Optimization → Backtesting & Evaluation

Test whether predictive skill translates into real investment value

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Methodology: Portfolio Strategies and Evaluation

Portfolio Strategies Compared

- Equal Weight (1/N)
- Global Minimum Variance (GMV)
- Hierarchical Risk Parity (HRP)
- Black–Litterman (BL)
- ML-based prediction portfolios

Evaluation Metrics

- Annualised return
- Annualised volatility
- Sharpe ratio
- Turnover
- Net performance after costs

Backtesting Framework

- Walk-forward expanding-window implementation
- Monthly portfolio rebalancing
- Transaction costs and turnover explicitly incorporated

Interpretation Logic

Prediction accuracy → Portfolio allocation
→ Risk-adjusted returns → Economic value

Final validation of whether the predictive structure identified in earlier phases is economically actionable

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: Portfolio Strategy Comparison

<i>Strategy</i>	<i>Return</i>	<i>Sharpe</i>	<i>Max DD</i>	<i>Turnover</i>
EqualWeight	17.50%	1.18	-9.20%	0.00%
GMV_EWMA	17.90%	1.49	-5.80%	5.80%
HRP_EWMA	17.40%	1.43	-5.80%	5.00%
MVO_GK	18.50%	1.2	-9.90%	44.00%
BL_GK	18.70%	0.81	-19.30%	22.80%
BL_Tactical_GK	19.00%	0.95	-15.80%	22.80%
MVO_Raw	23.20%	1.87	-4.70%	59.80%
MVO_Raw_IW	20.30%	0.99	-16.70%	25.90%

Table 9

Note: GMV = Global Min. Variance; HRP = Hierarchical Risk Parity; MVO = Mean-Variance Opt.; BL = Black-Litterman; GK = Grinold-Kahn rescaled views; Raw = unrescaled ML predictions; EWMA = exponentially weighted covariance; IW = inverse-volatility weights. Max DD = maximum drawdown. RF = 6.5% p.a.

- MVO_Raw achieved the best overall performance
 - Sharpe = 1.87
 - Annual Return = 23.2%
 - Lowest Max Drawdown among these strategies
- GMV_EWMA and HRP_EWMA also outperformed 1/N benchmark
- Black-Litterman strategies underperformed
- Raw prediction portfolios performed better than GK-scaled portfolios

Prediction-based portfolios generated economically meaningful alpha, especially when raw signals were used directly

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: Gross vs Net Performance

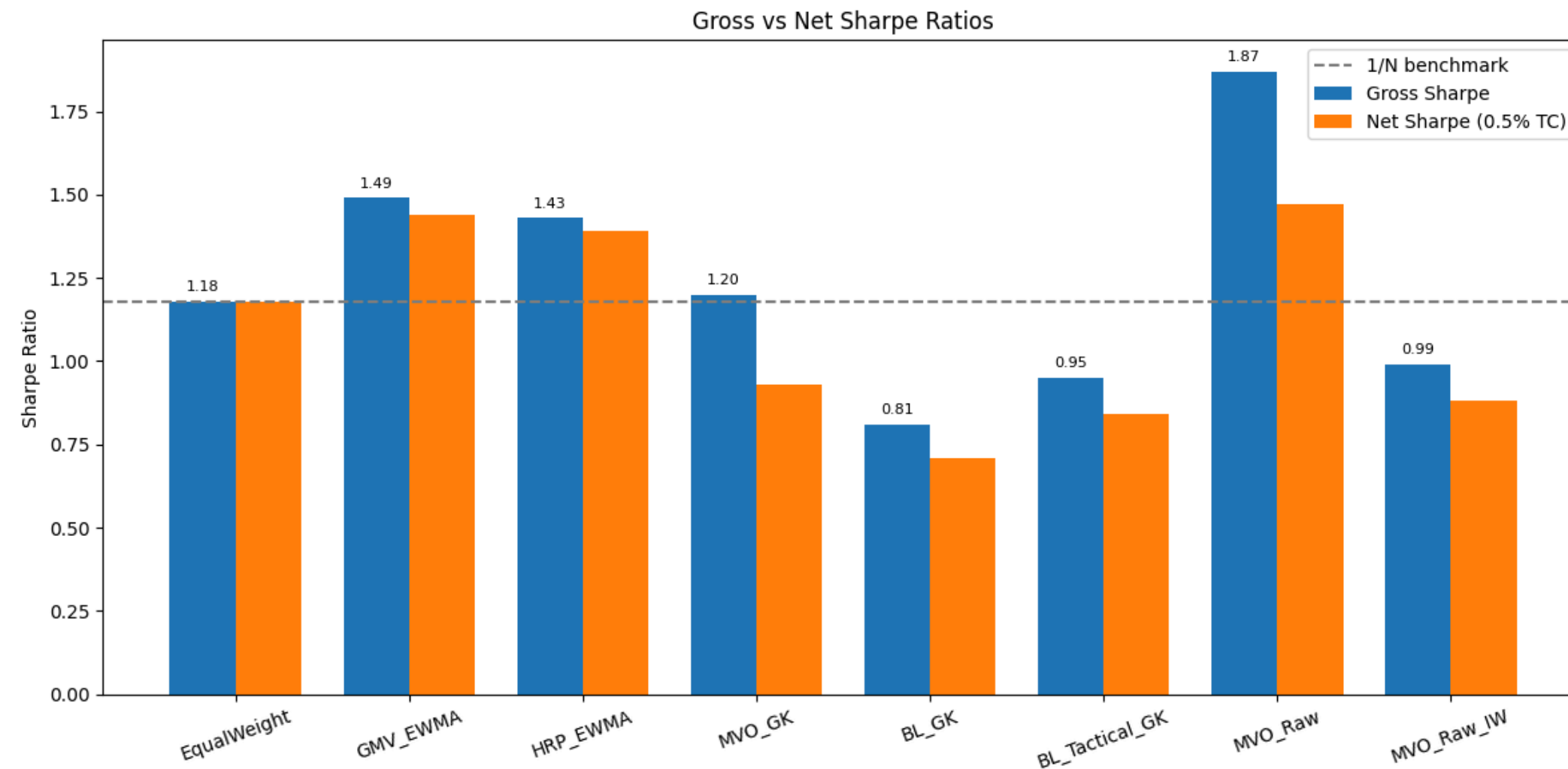


Fig.9

- MVO_Raw had highest gross Sharpe: 1.87
- After 0.5% transaction costs: Net Sharpe falls to 1.47
- Still remains superior to all other strategies
- High turnover is the main weakness: around 60% monthly turnover

Key Points

- Prediction quality alone is not enough
- Execution costs materially affect realized performance

The best-performing strategy is also the most sensitive to trading frictions

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: The Grinold–Kahn Ablation Result

<i>Strategy</i>	<i>Sharpe</i>	<i>Turnover</i>
MVO_GK	1.196	44.00%
MVO_Raw	1.868	59.80%

Table 10

- Both strategies use identical predictions
- Only difference:
 - MVO_GK applies GK scaling
 - MVO_Raw uses raw predictions
- GK scaling excessively shrinks strong signals
- In small universes (3 assets):
 - strong signals should create concentrated allocations
- Raw forecasts preserved predictive strength better

Grinold–Kahn scaling reduced performance substantially in this setting

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: Sensitivity to Transaction Costs

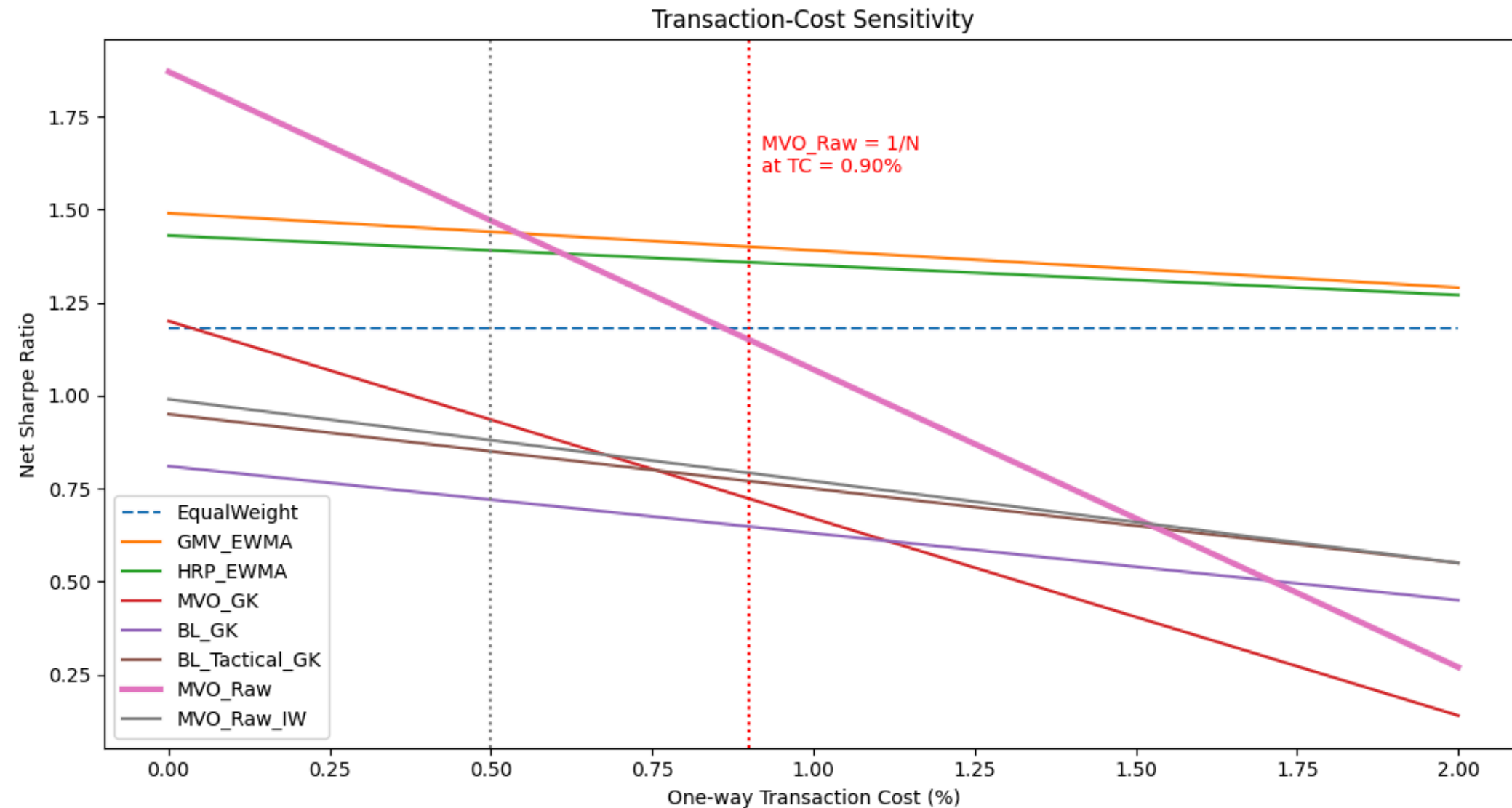


Fig.10

- MVO_Raw declines fastest as TC increases
- Break-even points:
 - vs 1/N → 0.90%
 - vs GMV → 0.55%
 - vs HRP → 0.65%
- At low TC:
 - prediction portfolios dominate
- At higher TC:
 - low-turnover risk-based strategies become preferable

Economic value exists, but depends critically on implementation efficiency and market liquidity

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: Stability Across Time

<i>Strategy</i>	<i>H1 Sharpe</i>	<i>H2 Sharpe</i>
MVO_Raw	1.72	1.99
GMV_EWMA	1.08	1.8
HRP_EWMA	1.03	1.75
BL_GK	1.14	0.43

Table 11

[H1 (December 2023 to December 2024) and H2 (January 2025 to February 2026)]

- MVO_Raw remained strong in both subperiods
- GMV and HRP improved sharply during REIT rally
- Black–Litterman weakened substantially in H2
- Suggests MVO_Raw performance is not driven by one isolated regime

Raw prediction portfolios showed the strongest temporal robustness

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: Bootstrap Validation and Portfolio Weights

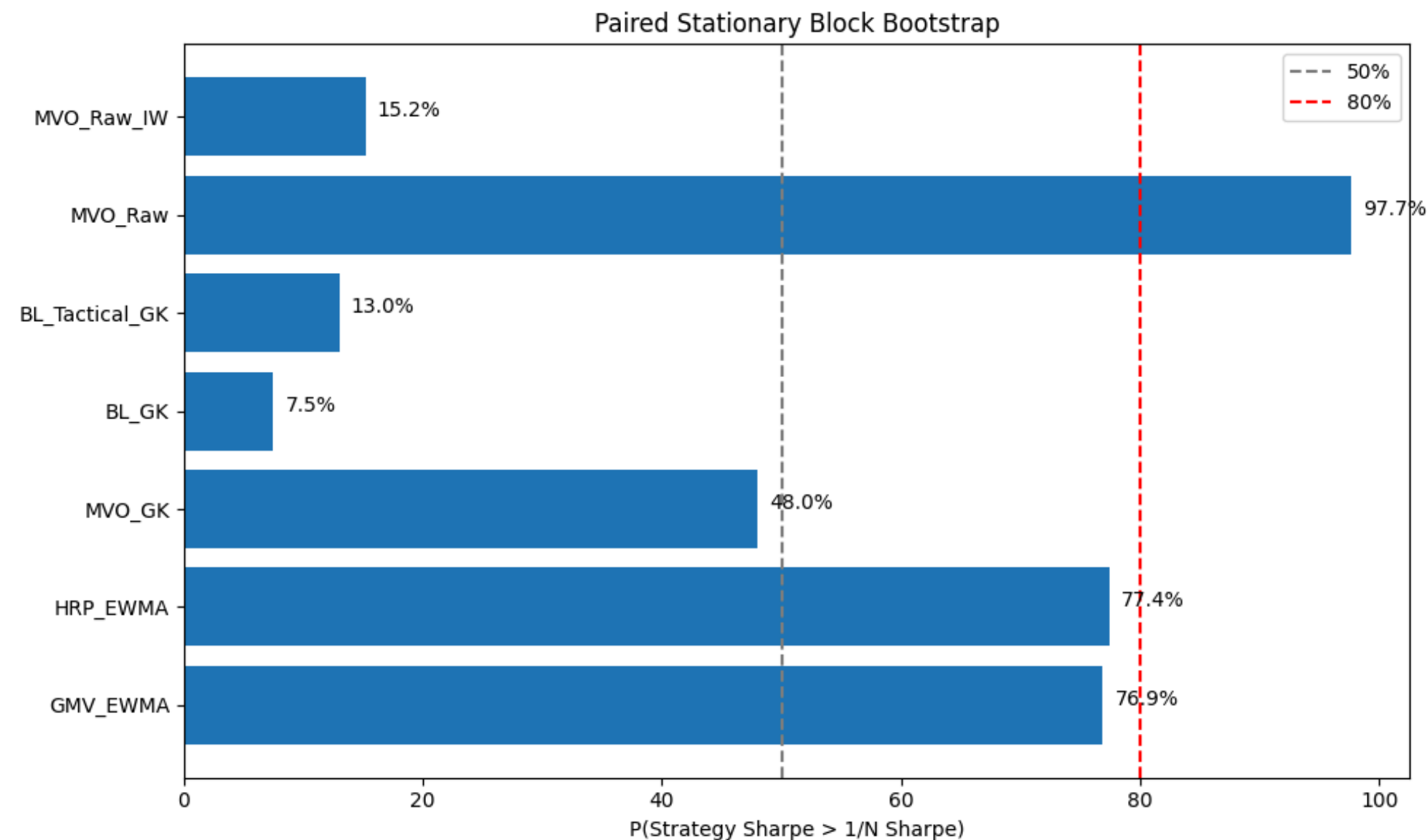


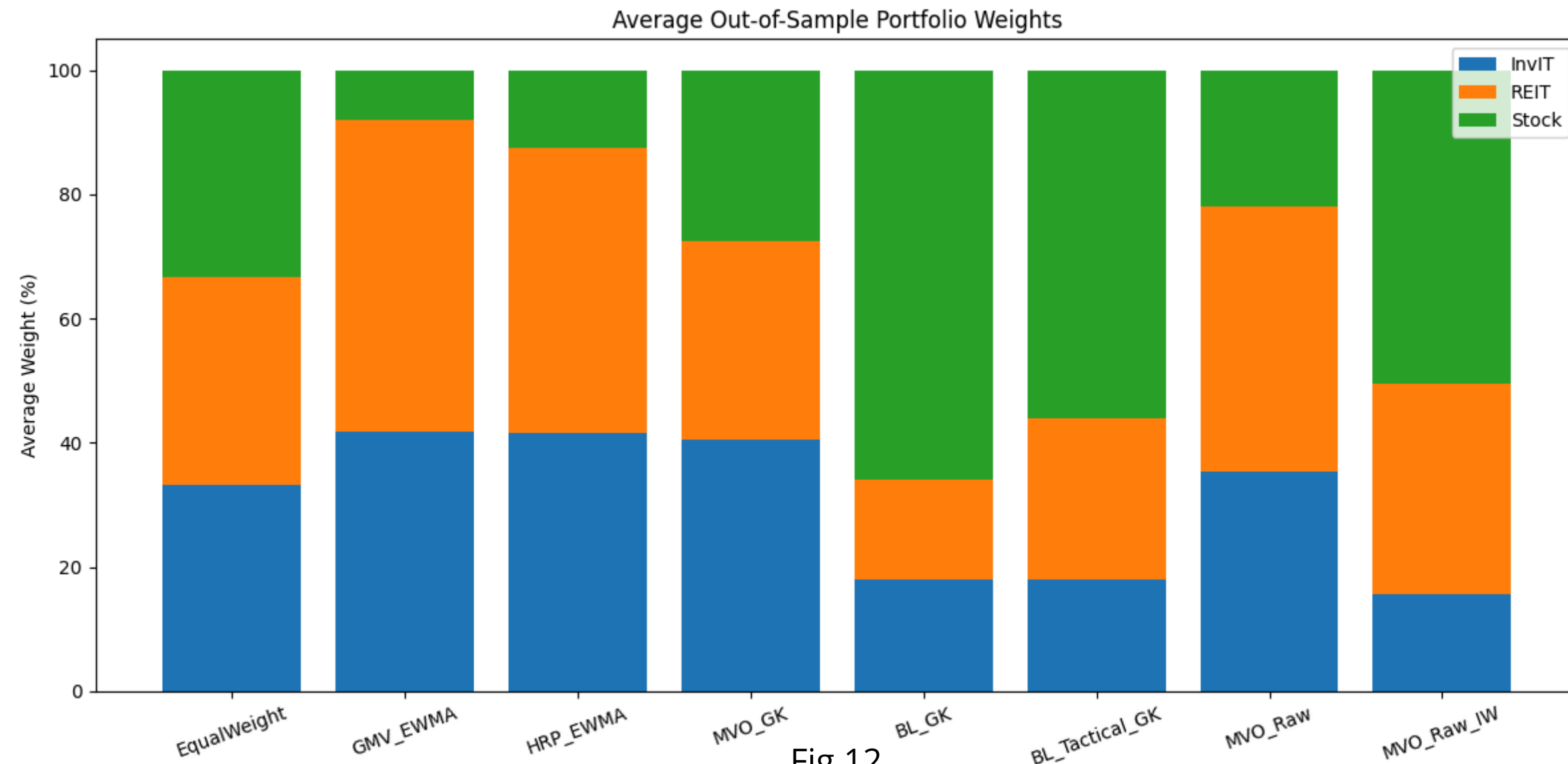
Fig.11

Bootstrap Results

- MVO_Raw: 97.7% probability of beating 1/N
- GMV and HRP: around 77% probability
- GK-based strategies: mostly below 50%

PHASE 4: PORTFOLIO OPTIMISATION AND ECONOMIC EVALUATION

Results: Portfolio Allocation Behaviour Across Strategies



GMV_EWMA and HRP_EWMA tilt toward lower-volatility REIT/InvIT assets, while Black-Litterman strategies allocate more heavily toward Stocks. MVO_Raw shows the strongest active positioning with balanced REIT-InvIT exposure and moderate Stock allocation

MANAGERIAL IMPACT

Indian Institutional Managers

- Prediction-based allocation (MVO_Raw) significantly outperformed equal-weight portfolios
- Transaction costs are critical: high-turnover strategies lose advantage quickly
- Illiquidity (Amihud) should be incorporated into InvIT position sizing

International Portfolio Managers

- Indian InvITs/REITs show low global integration, i.e, strong diversification benefits
- Suggested allocation range: 3–8% within global infrastructure portfolios
- INR depreciation risk can be managed through short-term currency hedging

Sponsors & Policy-Makers

- Lower-yield environments are more favourable for InvIT/REIT issuance
- Spillover indices can serve as macroprudential monitoring tools
- Gradual foreign participation liberalisation may reduce systemic instability

MANAGERIAL IMPACT

Practical Takeaways

- Low spillovers indicate Indian infrastructure assets remain largely domestically driven
- Machine learning signals add economic value when combined with portfolio optimisation
- Covariance-based methods (GMV/HRP) provide more stable performance under transaction costs

Limitations

- Results based on a relatively short 27-month out-of-sample window
- Findings may change under different macroeconomic regimes
- Conclusions are specific to the InvIT-REIT-Infrastructure Stock universe

Indian InvITs and REITs exhibit low global integration and strong domestic return structure, creating meaningful diversification and portfolio opportunities

CONCLUSION

Principal Findings:

- Indian InvITs and REITs exhibit strong segmentation from global markets
- foreign shocks explain only around 0.3% to 2.5% of variance
- A clear three-level factor structure emerges:
 - Stocks strongly macro-driven
 - REITs partially explained
 - InvITs largely idiosyncratic and liquidity-sensitive
- Bayesian Ridge with Phase 2 priors achieves strongest predictive performance
- Prediction-based portfolios generate economically meaningful value:
 - MVO_Raw Sharpe = 1.87 gross / 1.47 net
- Grinold–Kahn scaling reduces performance in small-asset settings with strong signals

CONCLUSION

Contributions:

- Establishes an empirical baseline for Indian InvIT and REIT markets
- Integrates econometrics, machine learning, and portfolio optimisation into a unified framework
- Demonstrates how Bayesian priors from factor models improve predictive modelling

Limitations:

- Relatively short out-of-sample period (27 months)
- Small asset universe (InvIT, REIT, Infrastructure Stock)
- Results sensitive to transaction costs and market regime changes
- Fixed DCC parameters used for stability

Future Research:

- Extend out-of-sample window over longer horizons
- Estimate DCC parameters directly using QMLE
- Apply TVP-VAR and high-frequency volatility frameworks
- Study impact of regulatory liberalisation on spillovers
- Explore Bayesian neural network priors

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THANK YOU